



Baicalin protects sertoli cells from heat stress-induced apoptosis via activation of the Fas/FasL pathway and Hsp72 expression

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ABSTRACT

Certain Chinese herbal medicines have antipyretic effects in both animal and human clinical practice. However, no report indicates their antipyretic effects on heat-stressed cells. The present study aimed to identify the protective effects of baicalin on the apoptosis of primary cultured bovine sertoli cells (SCs) subjected to heat stress (HS). The results demonstrated that HS induced apoptosis in the SCs exposed to 43 °C for 1 h as Fas/FasL was activated and caspase-3 was cleaved, the cells apoptotic rate was decreased. Moreover, the mRNA and protein levels of Hsp72 increased, whereas the cells apoptotic rate and expression of Fas, FasL, caspases 8 and 3 decreased in the SCs pretreated with various concentrations (0.1, 1, 10, 20 µg/mL) of baicalin prior to HS. In conclusion, baicalin ameliorates heat stress-induced cell apoptosis via the modulation of the cell survival rate through Fas/FasL pathway activation and the upregulation of Hsp72 expression in bovine SCs.

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1. Introduction

Normally, testes are maintained 2–8 °C below the core body temperature [1] for optimal spermatogenesis. Mild hyperthermia disturbs their function and ultimately inhibits spermatogenesis [2]. A study on mice has demonstrated that the occurrence of transient heat stress (HS) leads to testicular weight loss, increase cell death, loss of reproductive cells and fertility [3]. Because of testicular sensitivity to an increase in temperature, the body has different mechanisms to maintain the proper temperature for normal orchid function [4]. Sertoli cells (SCs), which are considered nursing cells for developing sperm [5], are somatic cells in the testes and play an important roles in the process of sperm production. These cells are also easily affected by heat stress, which renders them unable to perform a supportive role for germ cell development, and they impairs male fertility [6]. Normally, the testis has been demonstrated to express the Fas ligand (FasL) at the highest level among tissues, and FasL expression is exclusively restricted to SCs [7], which regulates the number of spermatogenic cells by binding Fas receptors on the spermatogenic cells membrane.

The Fas/FasL system has been proposed as a key regulator of the activation of germ cell apoptosis [8]. Both FasL and Fas, which are expressed by SCs and germ cells, respectively, respond to environmental conditions and initiate germ cell death. It is advisable to evaluate the expression kinetics of both FasL and Fas after the induction of massive germ cell death in various testicular injury models. Evidence from previous studies suggests a role of DNA damage in the heat stress-induced apoptosis of SCs [9]. If SCs and germ cells are unbalanced, the sperm production and quality will decrease [10].

Heat shock proteins (HSPs), which were originally identified as stress-responsive proteins, belong to the most prominent group of proteins with functions involved in the folding and unfolding of other proteins [11]. The 70-kDa heat shock proteins (Hsp70s), which comprise the major class of proteins induced by increased temperatures, are chaperones that assist in the folding, assembling and disassembling of protein complexes [12]. The Hsp70 family includes a 73-kDa protein (Hsc73) which is constitutively expressed in healthy cells and a highly stress-inducible 72-kDa protein (Hsp72) which is normally typically expressed at low levels in healthy cells [13]. Untreated SCs do not normally express Hsp72; however, heat stress induces Hsp72 expression in the cell cytoplasm [14]. It has been demonstrated that heat stresses trigger apoptosis in somatic cells and induce Hsp72 gene expression, which limits apoptosis [15].

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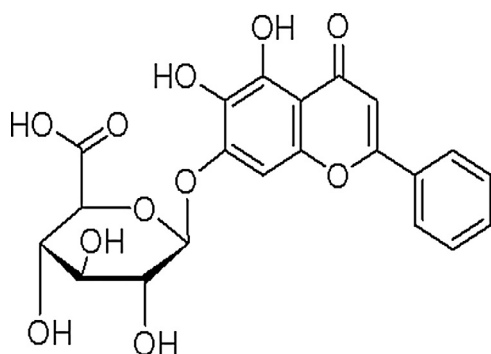


Fig. 1. Chemical structure of baicalin with a relative molecular weight of 446.35.

Baicalin, a type of flavonoid (Fig. 1), is extracted from the dried root of *Scutellaria baicalensis* Georgi [16], which is a traditional Chinese herbal medicine clinically used as an antipyretic, bacteriostatic and anti-inflammatory drug [17]. Studies have also demonstrated that baicalin can reduce the temperature of rats suffering with fever [18] and has pharmacological effects, including anti-oxidation activity [19] and the scavenging of oxygen-free radicals [20]. Baicalin effectively protects against concanavalin A-induced liver cell apoptosis [21]. In our previous study it was found that baicalin increased SCs survival rate under heat stress and improved expressions of both stem cell factors (SCF) and glial cell line-derived neurotrophic factor (GDNF) [22]. However, minimal information exists in the literature regarding the role of baicalin in SCs apoptosis induced by heat stress. This study aimed to address the hypothesis that baicalin can ameliorates heat stress-induced apoptosis through the Fas/FasL pathway and the regulation of Hsp72 expression in bovine SCs.

2. Materials and methods

2.1. Materials

Baicalin (concentration $\geq 98\%$) was bought from the National Pharmaceutical Engineering Center (Jiangxi, China). Dulbecco's Modified Eagle Media: Nutrient Mixture F-12 (DMEM/F-12), goat serum, penicillin and streptomycin for cell culture use were obtained from Life Technologies, Inc. (Grand Island, NY, USA). Collagenase Type IV, trypsin, dimethyl sulfoxide (DMSO) and CASP-3-C were obtained from Sigma Chemical Co. (St. Louis, MO, USA). Goat polyclonal anti-Hsp72 (sc-1060) and GAPDH (sc-48166) and rabbit polyclonal anti-Fas (sc-7886) and FasL (sc-834) antibodies were all obtained from Santa Cruz Biotechnology, Inc. (USA). Rabbit polyclonal anti-caspase-8 (ab-4052) and

caspase-3 (ab-90437) antibodies were obtained from Abcam (US), Inc. (UK). Secondary polyclonal anti-rabbit and anti-goat alkaline phosphatase-conjugated antibodies were purchased from Jackson ImmunoResearch Laboratories, Inc. (USA), Sigma.

2.2. Sertoli cells purification, culture and identification

Testes of bull calves were obtained from a calf serum-producing factory within 4 h after slaughter, the tunica albuginea was removed and seminiferous tubules were isolated from connective tissues under a stereomicroscope [23]. Collagenase IV (0.15%) was used to separated SCs from the tubules and remove the interstitial tissue in the supernatant; the remainder was subsequently digested in 0.25% trypsin. The suspension was filtered through a stainless steel 400-mesh sieve to remove undissociated tissues and debris; the filtrate was then pelleted by centrifugation at 1000 r/min for 10 min and washed twice.

Cell viability and density were determined by trypan blue exclusion using a hemocytometer [24]. The isolated cells were plated and cultured as dispersed single cells onto a tissue culture dish with a density of approximately 1×10^6 cells/mL in non-serum DMEM/F-12 medium and antibiotics at 34 °C under humidified atmosphere of 5% CO₂. After 5 h of incubation, majority of the cells spontaneously adhered to the surface of the flask. The non-adhering cells were removed, while the adhered cells were sequentially cultured with serum starvation for 48 h, digested with 0.25% trypsin and cultured with 10% fetal bovine serum (FBS) and 1% antibiotics DMEM/F-12 until the third generation for the experiment.

After purification, SCs were identified on the basis of their characteristic feature of bipolar corpuscle around the nucleus seen after Feulgen staining [25] and their unique expression of FasL via immunocytochemistry [26].

2.3. Cell treatment

Baicalin was dissolved in PBS and diluted with complete culture medium. In each experiment, the cells were incubated in the presence of baicalin (0.1, 1, 10 and 20 $\mu\text{g/mL}$, respectively) for the treated groups or the absence of baicalin (0 $\mu\text{g/mL}$) for the controls at 34 °C for 24 h prior to HS at 43 °C for 1 h and recovery at 34 °C for 6 h.

2.4. Identification of the cell apoptotic rate

SCs were washed with cold PBS and stained with propidium iodide (PI) and fluorescein isothiocyanate (FITC)-conjugated annexin V (FITC-AnxV) following the manufacturer's instructions. The cells were analyzed using BD FACS Calibur (USA).

Table 1
Primer sequences for the qPCR assay.

Gene	Primer sequences	PCR product size (bp)	GenBank [No.]
Hsp72	TGTCGCTGGGACTGGAGA GCTGGTTGTCCGAGTAGTG	111	NM_203322.2
FasL	ACTACCGCCACCACTCTGA GGCCACCAGAACCATGAAAA	85	NM_001098859.1
Fas	AAAAACTGGGGCTGCCCTTA CTTTGTGGGGATGGAACAA	148	NM_174662.2
Caspase-8	CTGAGAGAAGAGGCCCTGA CCCGGCTTAGGAACCTGAGG	173	DQ319070.1
Caspase-3	GAGCCTGTGAGCGTGCTTTT TGGTGCTGAGGATGACATGG	163	NM_001077840.1
GAPDH	ACGGCACAGTCAAGGCAGAG GTGATGGCGTGACAGTGTT	376	NM_001034034.2

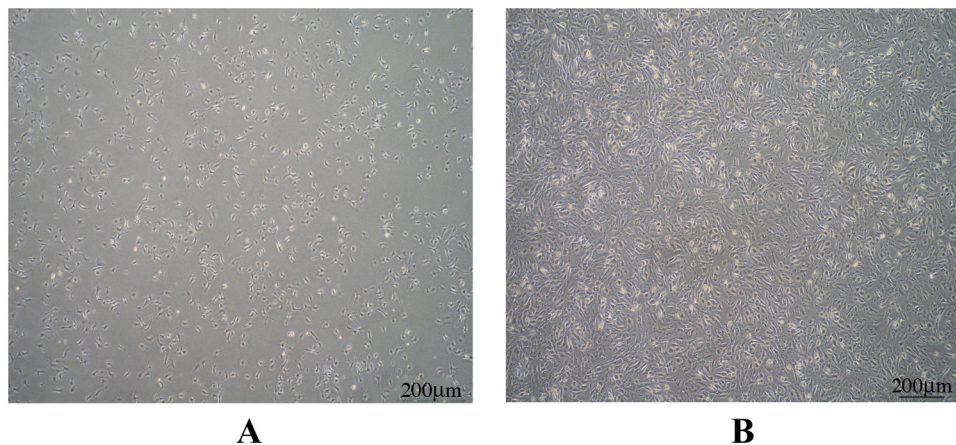


Fig. 2. Appearance of cultured calf sertoli cells (40×). (A) Sertoli cells inoculation after 5 h and (B) sertoli cells cultured at the logarithm.

2.5. RNA extraction and real-time quantitative PCR (qPCR)

Extraction of total RNA from SCs was performed using RN28-Easyspin Plus Tissue/Cell RNA Kit (Aidlab Biotech, Beijing, China) to extracted total RNA from SCs according to the instructions and template cDNA for real-time PCR was synthesized using Revert Aid First Strand cDNA Synthesis Kit (Thermo Fisher, USA) according to manufacturer's instructions. Quantitative real-time PCR was performed using SYBR Green I and the LightCycler 480® System (Roche Diagnostics, Basel, Switzerland). Briefly, 1.0 μL of cDNA was added to 5 μL of SYBR Green PCR Master Mix (Roche) and 0.2 μL of oligonucleotide primer (Sangon Biotech, Shanghai, China) (Table 1) to a final volume of 10 μL and was subjected to the following conditions: 95 °C for 10 min, followed by 45 cycles of 95 °C for 10 s, 60 °C for 20 s, and 72 °C for 30 s in 96-well optical reaction plates (Roche), the melting curves were analyzed at 65 °C and 97 °C after 45 cycles [27]. Each qPCR analysis was performed in triplicate. The relative mRNA concentrations were calculated based on the amplification efficiencies and the crossing point (CP) values [28].

2.6. Western blotting analysis

The samples for total protein were lysed in cell lysis buffer (Beyotime, China), and processed by SDS-PAGE (12%) gel electrophoresis; proteins in the gel were transferred to the PVDF membranes. The membranes were washed 3 times with TBST, blocked with 12% goat serum and left at room temperature for 2 h. They were then incubated with primary antibody against Hsp72 (1:1000 dilution), caspase 3 (1:1000 dilution), caspase 8 (1:1000 dilution), Fas (1:200 dilution), FasL (1:200 dilution) or GAPDH

(1:1000 dilution) at 4 °C. After washing and further incubation with secondary polyclonal anti-rabbit alkaline phosphatase-conjugated antibody (1:6000 dilution) and secondary polyclonal anti-goat alkaline phosphatase-conjugated antibody (1:4000 dilution) for 1 h at 25 °C, the blots were finally washed 3 times with TBST and twice with ultra pure water. Finally, an enhanced chemiluminescence (ECL) (TransGen Biotech, Beijing, China), was used to visualize the immunoreactive proteins.

2.7. Caspase 3 activity detection

The supernatant of the treated cells was collected and examined using the colorimetric Caspase-3 Assay Kit to detect caspase 3 activity according to the manufacturer's instructions.

2.8. Statistical analysis

All values are expressed as the mean of three replicates ± SD. Differences between groups were evaluated by one-way ANOVA followed by Dunnett's *t*-test using Graphpad Prism 5.01 (GraphPad Software, Inc.). A difference at $P < 0.05$ was considered statistically significant.

3. Results

3.1. Purity of SC preparations and identification

Separated primary SCs and cultured cells for the logarithm are shown in Fig. 2. The viability of the primary culture cell was 95% as determined by a trypan blue dye exclusion technique. Feulgen

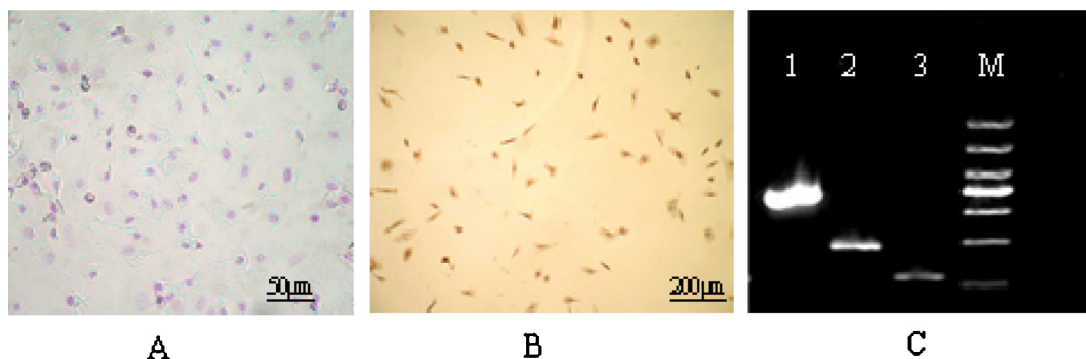


Fig. 3. Identification of sertoli cells: (A) sertoli cells nuclei with perinucleolar bodies (400×); (B) positive for expression of FasL (100×); (C) marker gene expression detection of sertoli cells, M: 1000 Maker; 1. GAPDH; 2. SCF; 3. GDNF.

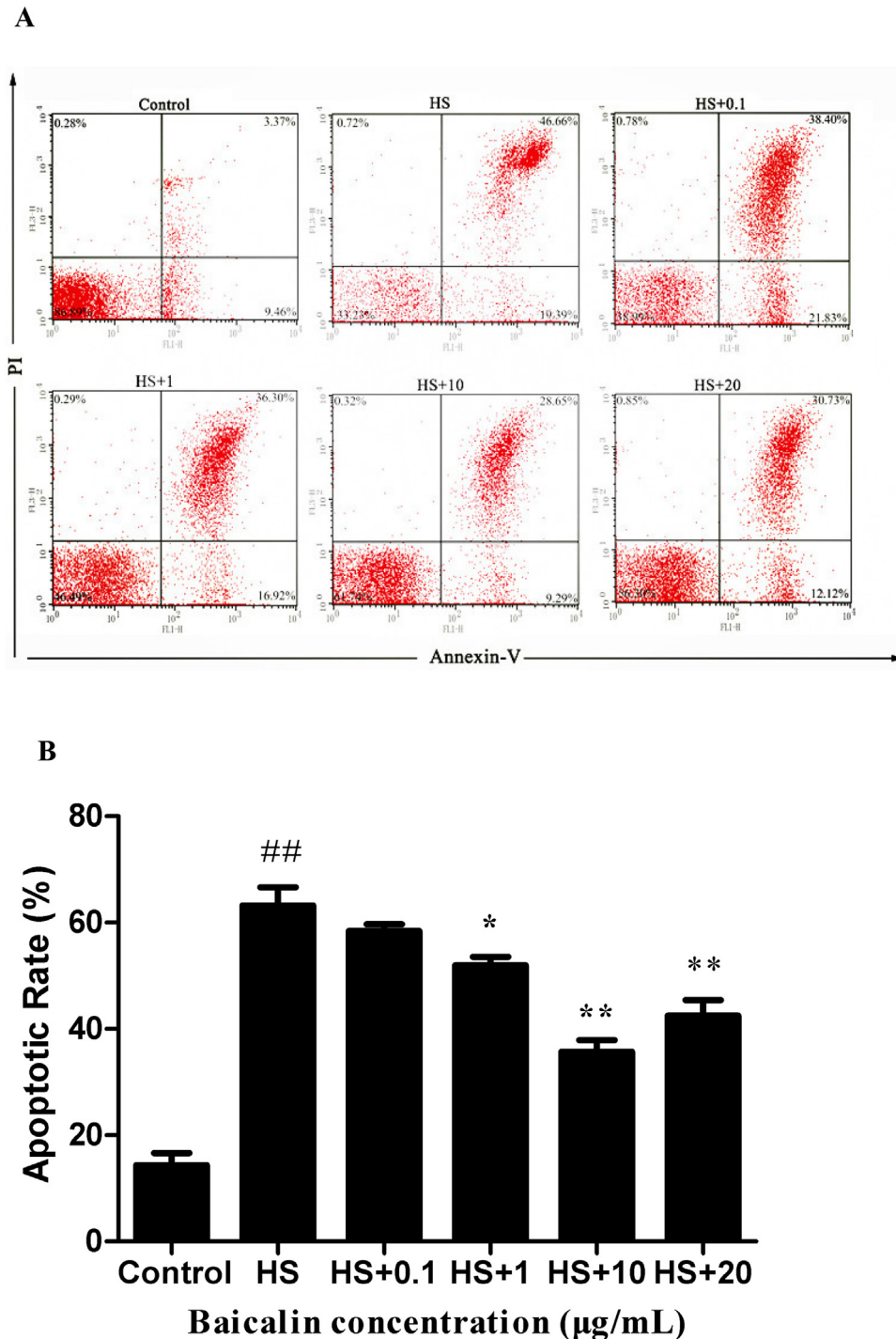


Fig. 4. Effects of HS with or without baicalin pretreatment on the apoptosis rate in sertoli cells. Cells were divided into following groups: control, heat stress (HS, 43 °C), HS+baicalin (0.1, 1, 10 and 20 μg/mL). Cells were staining with annexin-V and Plapoptosis analysis apoptosis by flow cytometry, the results The results drawn from A to B. #*p*<0.05 vs. control group; ##*p*<0.01 vs. control group. **p*<0.05 vs. HS group and ***p*<0.01 vs. HS group.

stained purified SCs with stained nuclei and unstained cytoplasm and satellite nucleosomes were on both sides of the nucleolus (Fig. 3A). The positive reaction of FasL showed positive reaction upon immunocytochemical identification (Fig. 3B).

3.2. Apoptotic effect of heat stress on sertoli cells

The results demonstrated an increase in the number of apoptotic cells in HS compared to the control group (*P*<0.01) (Fig. 4). HS activated Fas/FasL pathway, expressions of both Fas (Fig. 5A and B), FasL (Fig. 5C and D), caspase 8 (Fig. 6A and B) mRNA and proteins

were significantly increased (*P*<0.05, *P*<0.01) compared with the control group. Moreover, the caspase 3 was cleaved, the caspase 3 mRNA (Fig. 6C) and activity (Fig. 7) were significantly increased (*P*<0.01), while the procaspase 3 (Fig. 6D) protein was significantly decreased (*P*<0.01) compared to the control group.

3.3. Effect of baicalin on heat stress-induced apoptosis of sertoli cells

To investigate the protective effect of baicalin on sertoli cell apoptosis, which could resulted from heat stress, the cells were

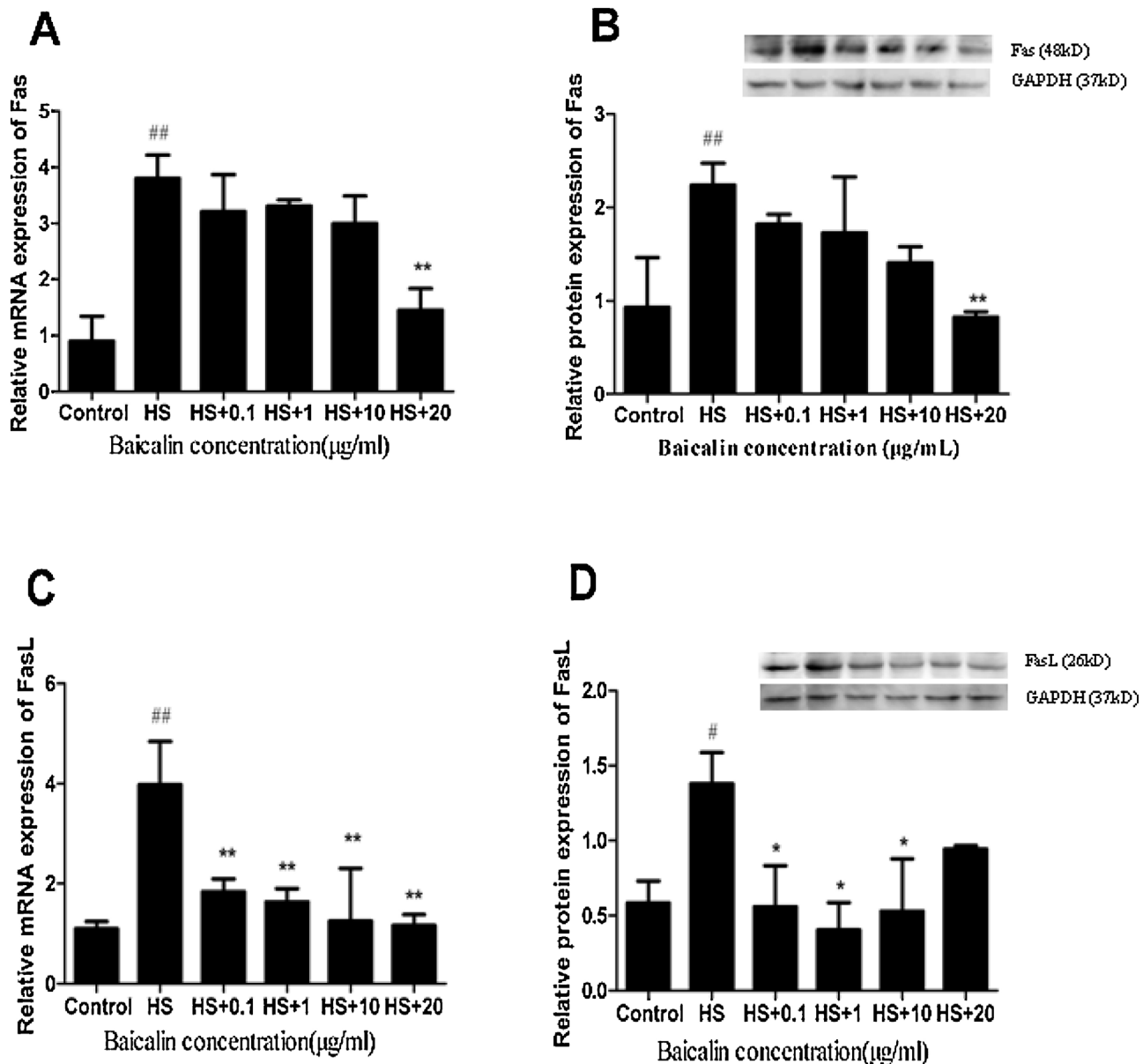


Fig. 5. Expressions of Fas and FasL mRNA (A and C) and their proteins (B and D). Cells were pretreated with baicalin and then subjected to HS, mRNA levels were analyzed by realtime PCR. Cells were divided into following groups: control, heat stress (HS, 43 °C), HS + baicalin (0.1, 1, 10, 20 μg). [#]*p* < 0.05 vs. control group; ^{##}*p* < 0.01 vs. control group. ^{*}*p* < 0.05 vs. HS group and ^{**}*p* < 0.01 vs. HS group.

analyzed via flow cytometry (Fig. 4A). The results demonstrated that pretreatment of SCs with 1, 10 or 20 μg/mL of baicalin significantly (*P* < 0.05, *P* < 0.01) decreased the number of apoptotic cells after heat stress treatment compared to cells subjected to HS only. Furthermore, the number of apoptotic cells decreased as the baicalin concentrations was increased (Fig. 4B).

Fas, FasL, caspases 8 and 3 expression levels were determined through qRT-PCR and Western blotting (Figs. 5 and 6). As shown in Fig. 5A and B, the Fas mRNA and protein levels in SCs treated with 20 μg/mL of baicalin were significantly decreased (*P* < 0.01) compared with the HS group. The FasL mRNA (Fig. 5C) levels in the cells treated with 0.1, 1, 10 and 20 μg/mL of baicalin were significantly decreased (*P* < 0.01), while the FasL protein (Fig. 5D) levels in the cells treated with 0.1, 1 and 10 μg/mL of baicalin also showed significantly decreased (*P* < 0.05) compared to the HS group.

Similarly, a significant decrease (*P* < 0.05, *P* < 0.01) in the caspase 8 mRNA (Fig. 6A) expression was observed in the cells treated with 1, 10 and 20 μg/mL of baicalin. However, the procaspase 8

protein (Fig. 6B) level in the cells treated with 10 and 20 μg/mL was significantly increased (*P* < 0.05, *P* < 0.01) compared to HS group.

In addition, the caspase 3 was activated, the caspase 3 mRNA (Fig. 6C) expression in SCs treated with 0.1, 1, 10 and 20 μg/mL of baicalin was significantly decreased (*P* < 0.05, *P* < 0.01), while the procaspase 3 protein (Fig. 6D) level in the cells treated with 10 and 20 μg/mL was significantly increased (*P* < 0.05, *P* < 0.01) compared to HS group. Moreover, a decrease in the caspase 3 activity (Fig. 7) in cells treated with 10 μg/mL (*P* < 0.05) and 20 μg/mL (*P* < 0.01) baicalin was recorded compared to the HS group.

3.4. Effects of baicalin on Hsp72 mRNA and protein levels

Hsp72 was determined via q-PCR and Western blotting (Fig. 8) to investigate the effect of baicalin treatment on Hsp72 expression. The results demonstrated that the expressions of both Hsp72 mRNA (Fig. 8A) and protein (Fig. 8B) were significantly increased (*P* < 0.05) after 1 h of HS compared with the control group. Moreover, the

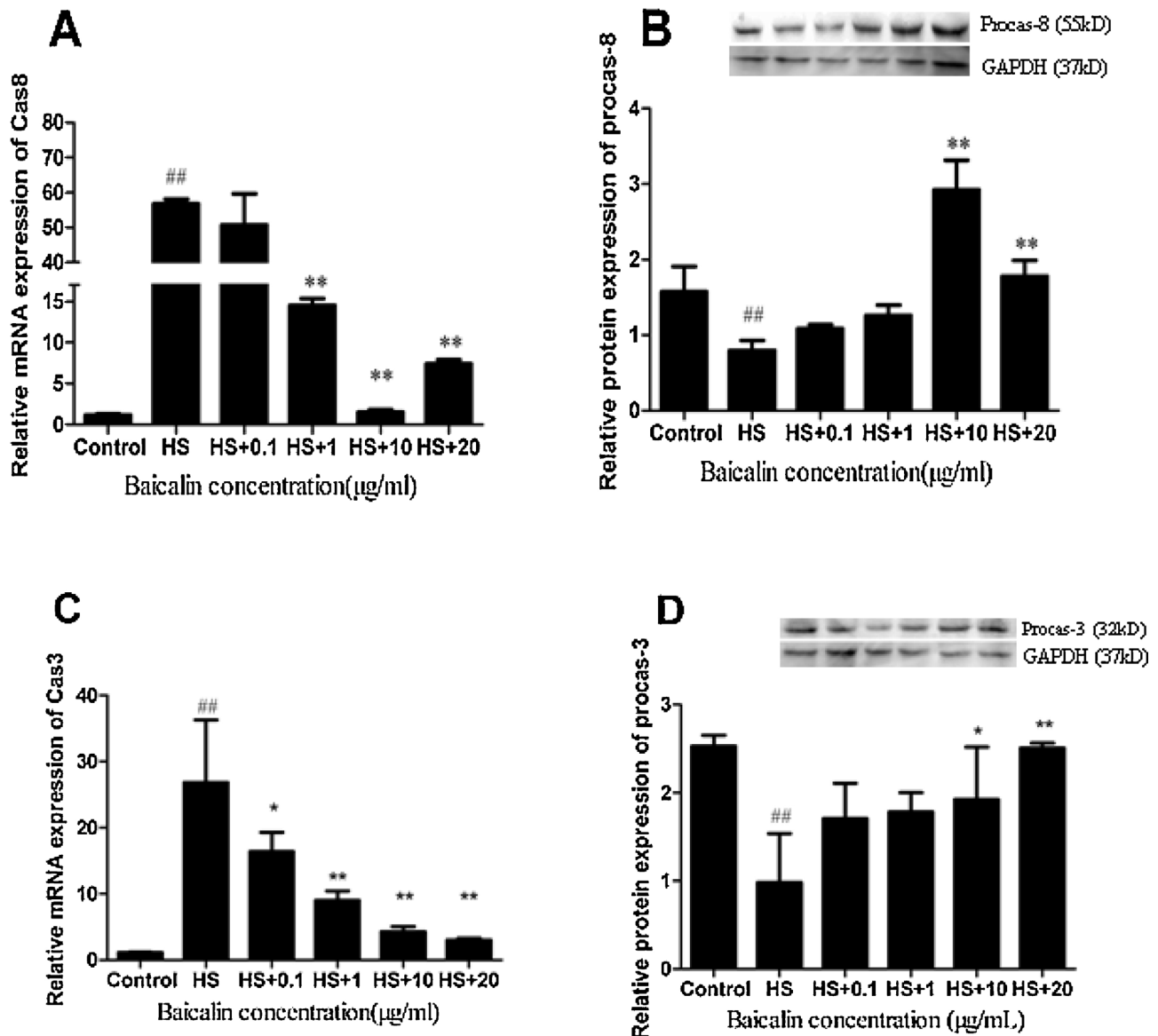


Fig. 6. Expression of mRNAs (A and C) and proteins (B and D) for caspase 8 and caspase 3. Cells were pretreated with baicalin and then subjected to HS, mRNA levels were analyzed by qRT-PCR. Cells were incubated with: control, heat stress (HS, 43 °C), HS + baicalin (0.1, 1, 10, 20 µg). ^{*}*p* < 0.05 vs. control group; ^{##}*p* < 0.01 vs. control group. ^{*}*p* < 0.05 vs. the HS group and ^{**}*p* < 0.01 vs. the HS group.

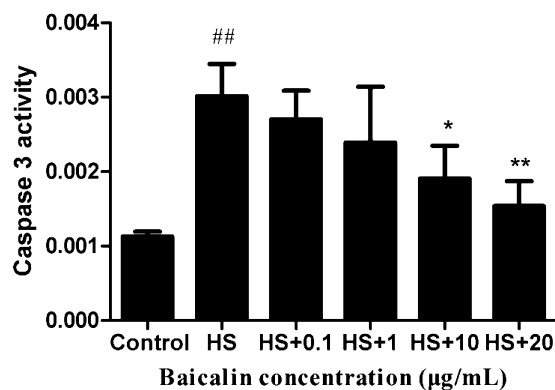


Fig. 7. Caspase 3 activity in all groups of sertoli cells. Cells were incubated with: control, heat stress (HS, 43 °C), HS + baicalin (0.1, 1, 10, 20 µg). ^{*}*p* < 0.05 vs. control group; ^{##}*p* < 0.01 vs. control group. ^{*}*p* < 0.05 vs. control group; ^{##}*p* < 0.01 vs. control group. ^{*}*p* < 0.05 vs. the HS group and ^{**}*p* < 0.01 vs. the HS group.

Hsp72 mRNA levels in the cells treated with 1 µg/mL of baicalin and the Hsp72 protein levels in the cells treated with 10 µg/mL of baicalin also significantly increased (*P* < 0.01) compared with the HS group.

4. Discussion

Testicular thermo regulation is important to maintain testicular temperature within an optimal range for spermatogenesis. Imao M. demonstrated that a hyperthermic challenge induced apoptosis in young mice and rats [30]. The rat testis tissue exposed to heat stress exhibited wider spaces in both the sertoli and spermatid cells [31]. SCs form a blood–testis barrier that creates a suitable microenvironment for development of germ cells under strict hormonal control [32]. The present study demonstrated that heat stress on SCs resulted in the induction of apoptosis. Nezhad et al. [9] also demonstrated that heat stress caused DNA damage and induced apoptosis in SCs, and cells were unable to provide perform a supportive role for germ cell development [33]. Furthermore, the

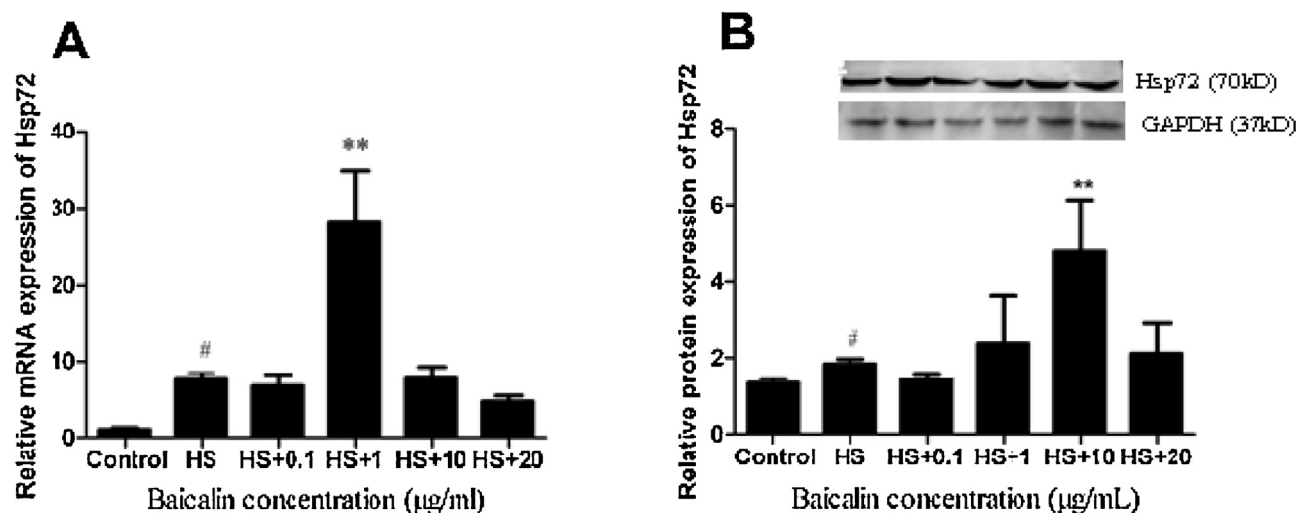


Fig. 8. Expression of Hsp72 mRNA (A) and its protein (B) in sertoli cells. Cells were pretreated with baicalin and then subjected to HS. Cells were divided into following groups: control, heat stress (HS, 43 °C), HS + baicalin (0.1, 1, 10, 20 μg). [#]*p* < 0.05 vs. control group; ^{**}*p* < 0.01 vs. control group. **p* < 0.05 vs. the HS group and ^{**}*p* < 0.01 vs. the HS group.

activation of Fas/FasL by heat stress may be involved in the apoptosis of sertoli cells because our results indicated that the expression of Fas, FasL and caspase 8 was increased, procaspases 8 and 3 were cleaved, and caspase-3 was activated. Apoptosis depends on the activation of caspases that will subsequently cleave a number of substrates [34]. Activation of Fas system also activates caspase 8, which in turn will further activate additional caspase 8 molecules, as well as downstream caspases, such as caspase 3 [35]. Therefore, we suggest that the apoptotic effect of heat stress on SCs is related to the induction of caspase activation.

Apoptosis is an important regulatory event in testicular homeostasis and the optimization of sperm production [36]. Therefore, the disruption of the apoptotic pathway and inhibition of apoptosis have become novel targets for cellular survival under heat stress. Baicalin, the main ingredient of a traditional Chinese herbal medicine, has been reported to have significant antipyretic effects and prevents cellular apoptosis [37]. Ye et al. [38] also demonstrated that baicalin can inhibit the TLR4 signaling pathway in a rat fever model and produce antipyretic effects.

The present study demonstrated that baicalin reversed the inhibitory effect of heat stress-induced SC apoptosis in a dose-dependent manner. Sun et al. [39] also demonstrated that baicalin can protect heat-stressed LLC-PK cells and decrease apoptosis rates at certain concentrations. Furthermore, we demonstrated that baicalin was effective in protecting SCs from heat stress-induced apoptosis not only because it suppressed the generation of Fas and FasL but also because it attenuated caspase-3 activity and upregulated procaspases 8 and 3 protein expression. Other studies have demonstrated that baicalin significantly suppresses caspase-3 in ischemic *gerbil hippocampi* reduces hippocampal apoptosis [40], and also protects cardiomyocytes from endoplasmic reticulum stress-induced apoptosis [41].

Hsps are capable of protecting cells from the harmful effects of stress by repairing denatured proteins or promoting damaged protein degradation [42], and they play an important role in apoptosis, resulting in mitochondrially mediated aging and male infertility. It has been demonstrated that the conditioning of cells with mild heat stress induces Hsp70 gene expression, which limits cellular apoptosis [15]. Hsp72 increases cell survival in response to apoptotic stimuli [43]. In our study, heat stress-induced Hsp72 expression increased, and treatment with baicalin significantly increased Hsp72 expression, whereas the caspase 3 activity decreased and the cell apoptosis rate significantly decreased. Experiments have

demonstrated that Hsp70 can block caspase-independent pathways to prevent cell death in conditions in which caspase activation does not occur [44]. Treatment with baicalin promoted Hsp70 expression and further resisted apoptosis in focal ischemia [45]. Our results suggest that HS induced Hsp expression, which can facilitate SC survival in stress conditions, and baicalin promoted Hsp72 expression and prevented the cells from resisting apoptosis. However, there is no direct evidence from the present study to determine whether baicalin regulates Hsp72 directly or jointly with the Hsp72 regulatory mechanism of apoptosis.

In summary, this study indicates that heat stress can induce SCs apoptosis through Fas/FasL pathway. These findings contribute to previously published data which demonstrate that baicalin is a promising anti-apoptotic agent that attenuates heat stress-induced cell apoptosis via the modulation of the expression of the activated Fas/FasL system and Hsp72 in a dose-dependent manner. This experiment provides a primary theoretical basis for the use of baicalin to improve reproductive performance in male animals exposed to high ambient temperature. However, considering the limitations of in vitro study, such as complicated crosstalk among different signaling pathways, it is not possible to clearly demonstrate whether baicalin can pass the blood–testis barrier and whether its effects on SCs also affect spermatogenesis. Additional investigations are necessary to confirm the effect of baicalin in vivo.

Conflict of interest

The authors declare that there are no conflicts of interest.

Transparency document

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